# 50 Special Question For Half yearly exam

(1) $f(x) = \sqrt{9-x^2}$ find domain & Range

(2) Draw graph of following function

a) motulus function

b) greatest integer function

c) signum function

(3) $f(x) = \dfrac{x^2}{1+x^2}$ find Range

(4) find

a) $\sin 15^\circ$

b) $\tan 75^\circ$

c) $\tan 22.5^\circ$

d) $\cot 15^\circ$

(5) To prove

$$\cos 4x = 1 - 8\sin^2 x \cos^2 x$$

(6) To prove.

$$\tan 4x = \frac{4\tan x\,(1-\tan^2 x)}{1 - 6\tan^2 x + \tan^4 x}$$

(7) Find number of words with or without meaning which can be made using all the letter of word AGAIN. If these words are written in a dictionary. what will be the 50th word

(8) How many numbers greater than 1000000 can be formed by using the digit 1, 2, 0, 2, 4, 2, 4 ?

(9) In how many ways can the letters of word PERMUTATIONS be arranged

(i) word start with P and end with S

(ii) Vowels are all together

(iii) There are always 4 letter between P and S

(10) Find number of different 8-letter arrangement can be form the letter DAUGHTER so that

(i) All vowels accur together

(ii) all vowels to not accur together

(11) If $\alpha$ and $\beta$ are different complex numbers with $|\beta| = 1$ then find

$$\left| \frac{\beta - \alpha}{1 - \bar{\alpha}\beta} \right|$$

(12) If $\left(\frac{1+i}{1-i}\right)^m = 1$ then find least value of m

(13) If $x - iy = \sqrt{\frac{a-ib}{c-id}}$ prove $(x^2+y^2)^2 = \frac{a^2+b^2}{c^2+d^2}$

(14) find conjugate of $\frac{(3-2i)(2+3i)}{(1+2i)(2-i)}$

(15) a) $\left[ i^{18} + \left(\frac{1}{i}\right)^{25} \right]^3$ find in $a+ib$

b) $\frac{5+\sqrt{2}i}{1-\sqrt{2}i}$

c) $\frac{(3+i\sqrt{5})(3-i\sqrt{5})}{(\sqrt{3}+\sqrt{2}i)-(\sqrt{3}-i\sqrt{2})}$

(16) find n $\frac{{}^nP_4}{{}^{n-1}P_4} = \frac{5}{3}$

(17) <u>To prove</u> $\cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) + \cos^2\left(x - \frac{\pi}{3}\right) = \frac{3}{2}$

(18) To prove $\sin 3x + \sin 2x - \sin x = 4 \sin x \cdot \cos\frac{x}{2} \cos\frac{3x}{2}$

(19) find $\sin\frac{x}{2}$, $\cos\frac{x}{2}$, $\tan\frac{x}{2}$

$\tan x = \frac{3}{4}$ $\quad \pi < x < \frac{3\pi}{2}$

(20) To prove

$\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x = 1$

(21) Prove

$\cos 2x \cos\frac{x}{2} - \cos 3x \cos\frac{9x}{2} = \sin 5x \cdot \sin\frac{5x}{2}$

(22) Find all pairs of consecutive odd natural number, both of which are larger than 10, such that their sum is less than 40

(23) find Range of each of

$f(x) = 2-3x$

$f(x) = x^2+2$

(24) How many words with or without meaning each of 3 vowels and 2 consonants can be formed from letters of $\boxed{\text{INVOLUTE}}$

(25) In how many ways letter of word $\boxed{\text{ASSASSINATION}}$ be arranged so that all are $\boxed{S}$ together.

(26) $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$ find $x$

(27) $\frac{2x-1}{3} \geq \left(\frac{3x-2}{4}\right) - \left(\frac{2-x}{5}\right)$ solve for real $x$

(28) solve system of linear inequalities

$3x - 7 < 5 + x$

$11 - 5x \leq 1$

To prove

(29) $\cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right) = -\sqrt{2}\sin x$

(30) To prove

$$\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

(31) $A = \{1, 2, 3, 4, 6\}$ let $R$ be relation on $A$ defined by

$\{(a, b) : a, b \in A$, $b$ is exactly divisible by $a$

(i) write $R$ in roster form

(ii) find domain of $R$

(iii) find Range of $R$

(32) If $A \cup B = A \cup C$ and $A \cap B = A \cap C$

Show $B = C$

(33) $U = \{1, 2, 3, 4, 5, 6\}$ $A = \{2, 3\}$

$B = \{3, 4, 5\}$

Prove demorgan law by above example

(34) $A = \{1, 2, \{3, 4\}, 5\}$ which are incorrect & why

(i) $\{3, 4\} \in A$ (ii) $\{\{3, 4\}\} \subset A$

(iii) $1 \in A$ (iv) $1 \subset A$ (v) $\{1, 2, 5\} \subset A$

(vi) $\phi \subset A$ (viii) $\{1, 2, 3\} \subset A$ (ix) $\phi \in A$

(35) Draw Venn diagram of following

(i) $(A \cup B)'$ (ii) $(A \cap B)'$

(36) To prove

$$(\cos x + \cos y)^2 + (\sin x - \sin y)^2 = 4\cos^2 \frac{x+y}{2}$$

(37) Prove $2\sin^2 \frac{3\pi}{4} + 2\cos^2 \frac{\pi}{4} + 2\sec^2 \frac{\pi}{3} = 10$

(38) If in two circle arcs of same length subted angles $60^\circ$ and $75^\circ$ at centre find ratio of their radii

(39) $2\,{}^nC_3 : {}^nC_3 = 12 : 1$ find $n$

(40) what is number of ways of chosing 4 cards from a pack of 52 playing cards.

(i) four cards are of same suit
(ii) four cards belong to 4 different suits
(iii) are face cards
(iv) two are red card and 2 are black cards
(v) cards are of same colour

(41) to prove $\cos 6x = 32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1$

(42) show
$\tan 3x \cdot \tan 2x \cdot \tan x = \tan 3x - \tan 2x - \tan x$

(43) In how many ways can 5 girls and 3 boys be seated in a row so that no 2 boys are together

(44) Determine the number of 5 cards combinations out of 52 card if selection of 5 cards has exactly one king.

(45) find a) $\operatorname{cosec}(-1410^\circ)$ b) $\tan \frac{19\pi}{3}$

c) $\cot\left(-\frac{15\pi}{4}\right)$ d) $\cos(-1710^\circ)$

(46) write set $\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}\}$

(47) The English alphabet has 5 vowels and 21 consonant. How many words with 2 different vowels and 2 different consonants can be formed from alphabet

(48) A manufacturer has 600 litres of 12% solution of acid. How many litres of 30% acid solution must be added to it so that acid content in mixture will be more than 15% but less than 18%.

(49) The longest side of A is 3 times the shortest side and third side is 2 cm shorter than the longest side. If perimeter of A is atleast 61 cm find minumum length of shorter side

(50) How many litres of water will have to be added to 1125 litres of 45% solution of acid so that resulting mixture will contain more than 25% but less than 30% acid content